

A nonzero class above the predicted top degree

Disproof of conjecture 00000001649

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Conjecture 00000001649 predicts that the largest degree of nonzero rational cohomology of $SL_n(\mathbb{Z})$ is

$$t(n) = \left\lfloor \frac{n^2 - 1}{4} \right\rfloor - 1.$$

There is no exclusion of $n = 2$ in either language version. For this standard case, integer arithmetic gives

$$t(2) = \left\lfloor \frac{3}{4} \right\rfloor - 1 = -1.$$

We exhibit a genuine nonzero cohomology class in degree zero, which lies strictly above this predicted top degree.

The actual group and its cochains. Take $G = SL_2(\mathbb{Z}) = \{M \in M_2(\mathbb{Z}) : \det M = 1\}$, with ordinary matrix multiplication, and trivial rational coefficients. In its inhomogeneous group cochain complex, $C^0 = \mathbb{Q}$, $C^1 = \text{Map}(G, \mathbb{Q})$, and the differential is

$$(d^0 q)(h) = h \cdot q - q = q - q = 0.$$

Thus $\ker d^0 = \mathbb{Q}$. Since there are no negative-degree cochains, $\text{im } d^{-1} = 0$, so

$$H^0(SL_2(\mathbb{Z}); \mathbb{Q}) = \ker d^0 / \text{im } d^{-1} = \mathbb{Q}/0 \cong \mathbb{Q}.$$

The class of the constant zero-cochain 1 is nonzero: its image under this isomorphism is $1 \neq 0$. In particular, H^0 has dimension one.

Contradiction. By the conjecture's definition, a largest nonzero degree d must satisfy $i \leq d$ whenever $H^i(SL_2(\mathbb{Z}); \mathbb{Q}) \neq 0$. The class above therefore forces $0 \leq d$. The proposed value $d = t(2) = -1$ instead forces $0 \leq -1$, a contradiction. This argument needs no calculation of higher cohomology and no assumption about its vanishing. A single nonzero class above a proposed maximum refutes that maximum.

Scope and arithmetic. The subtraction in the displayed conjecture is ordinary integer subtraction; $\lfloor 3/4 \rfloor - 1 = -1$, rather than natural-number truncated subtraction. The asserted formula is already false at $n = 2$, independently of the additional virtual-cohomological-dimension and boundary descriptions. An amended formula restricted to larger n would be a different statement.

Lean certificate. The project uses Mathlib's actual 2×2 integer special linear group, constructs trivial-action group cochains, the degree-zero differential, cocycles and the zero-boundary quotient, and gives an explicit linear equivalence with $\mathbb{Q}$. It certifies a nonzero class, dimension one, the integer floor computation and the impossibility of any proposed top degree equal to $t(2)$ satisfying even its necessary degree-zero bound.